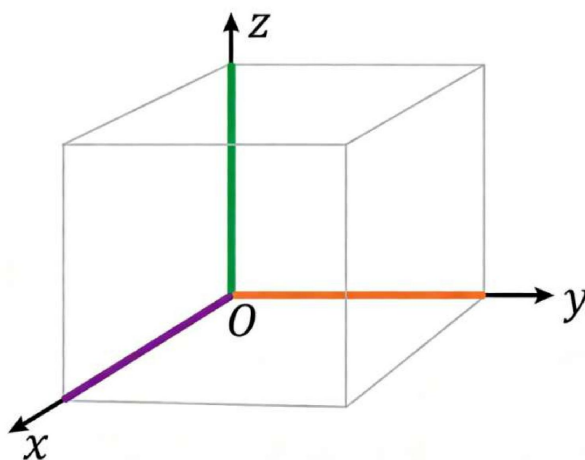


空间直角坐标系综合习题 (中英双语版)

一、基础练习 (Basic Exercises)

第 1 题 (Question 1): 在空间直角坐标系中, 尝试标出下列各点的位置: (In the space rectangular coordinate system, try to mark the positions of the following points: $A(0,2,4)$, $B(1,0,5)$, $C(0,2,0)$, $D(1,3,4)$.)



第 2 题 (Question 2): 在空间直角坐标系 $Oxyz$ 中, 哪个坐标平面与 x 轴垂直? 哪个坐标平面与 y 轴垂直? 哪个坐标平面与 z 轴垂直? (In the space rectangular coordinate system $Oxyz$, which coordinate plane is perpendicular to the x -axis? Which coordinate plane is perpendicular to the y -axis? Which coordinate plane is perpendicular to the z -axis?)

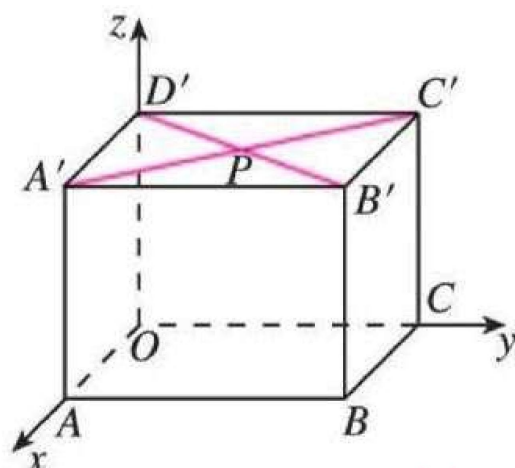
第 3 题 (Question 3): 写出点 $P(2,3,4)$ 在三个坐标平面内的射影的坐标。

(Write the coordinates of the projection of point $P(2,3,4)$ in the three coordinate planes.)

第 4 题 (Question 4): 写出点 $P(1,3,5)$ 关于原点成中心对称的点的坐标。

(Write the coordinates of the point that is centrally symmetric to point $P(1,3,5)$ with respect to the origin.)

第 5 题 (Question 5): 在长方体 $OABC - D'A'B'C'$ 中, $OA = 3$, $OC = 4$, $OD' = 3$, $A'C'$ 与 $B'D'$ 相交于点 P , 建立空间直角坐标系 $Oxyz$ 。



5.1 写出点 C 、 B' 、 P 的坐标。(In the cuboid $OABC - D'A'B'C'$, $OA = 3$, $OC = 4$, $OD' = 3$, and $A'C'$ intersects $B'D'$ at point P . Establish a space rectangular coordinate system $Oxyz$ and write the coordinates of points C 、 B' 、 P .)

5.2 写出向量 $\overrightarrow{BB'}$ 、 $\vec{A'C'}$ 的坐标。(Following Question 5, write the coordinates of the vectors $\overrightarrow{BB'}$ and $\vec{A'C'}$.)

第 6 题 (Question 6): 已知点 B 是点 $A(3,4,5)$ 在坐标平面 Oxy 内的射影, 求 $|\overrightarrow{OB}|$ (即线段 OB 的长度)。 (It is known that point B is the projection of point $A(3,4,5)$ in the coordinate plane Oxy . Find $|\overrightarrow{OB}|$ (i.e., the length of line segment OB).)

二、选择题 (Multiple Choice Questions)

第 1 题 (Question 1): 下列关于右手直角坐标系的判断方法, 正确的是 ()

(Which of the following is correct about the judgment method of the right-handed rectangular coordinate system?)

A. 右手拇指指向 y 轴正方向, 食指指向 x 轴正方向, (www.sicyes.com) 中指指向 z 轴正方向 (The thumb of the right hand points to the positive direction of the y -axis, the index finger points to the positive direction of the x -axis, and the middle finger points to the positive direction of the z -axis)

B. 右手拇指指向 x 轴正方向, 食指指向 z 轴正方向, 中指指向 y 轴正方向 (The thumb of the right hand points to the positive direction of the x -axis, the index finger points to the positive direction of the z -axis, and the middle finger points to the positive direction of the y -axis)

C. 右手拇指指向 x 轴正方向, 食指指向 y 轴正方向, 中指指向 z 轴正方向 (The thumb of the right hand points to the positive direction of the x -axis, the index finger points to the positive direction of the y -axis, and the middle finger points to the positive direction of the z -axis)

D. 右手拇指指向 z 轴正方向, 食指指向 y 轴正方向, 中指指向 x 轴正方向 (The thumb of the right hand points to the positive direction of the z -axis, the index finger points to the positive direction of the y -axis, and the middle finger points to the positive direction of the x -axis)

第 2 题 (Question 2): 已知空间中两点 $M(2, -1, 3)$ 和 $N(4, 3, -1)$, 则向量 $\overrightarrow{MN}$ 的坐标为 () (Given two points $M(2, -1, 3)$ and $N(4, 3, -1)$ in space, the coordinate of vector $\overrightarrow{MN}$ is ())

- A. $(2, 4, -4)$ B. $(-2, -4, 4)$ C. $(6, 2, 2)$ D. $(1, 1, 1)$

第 3 题 (Question 3): 空间直角坐标系中, Oyz 平面是由哪两条坐标轴确定的? () (In the space rectangular coordinate system, which two coordinate axes determine the Oyz plane? ())

- A. x 轴和 y 轴 (x -axis and y -axis)
B. y 轴和 z 轴 (y -axis and z -axis)
C. x 轴和 z 轴 (x -axis and z -axis)
D. 以上都不对 (None of the above)

第 4 题 (Question 4): 关于空间 (www.sicyes.com) 中点 $P(x, y, z)$ 的坐标几何意义, 下列说法正确的是 () (Regarding the geometric significance of the coordinates of point $P(x, y, z)$ in space, which of the following is correct? ())

- A. x 是点 P 在 y 轴上的射影坐标 (x is the projection coordinate of point P on the y -axis)
B. y 是点 P 在 z 轴上的射影坐标 (y is the projection coordinate of point P on the z -axis)
C. z 是点 P 在 z 轴上的射影坐标 (z is the projection coordinate of point P on the z -axis)
D. 坐标 (x, y, z) 与向量 $\overrightarrow{OP}$ 的坐标无关 (The coordinate (x, y, z) has nothing to do with the coordinate of vector $\overrightarrow{OP}$)